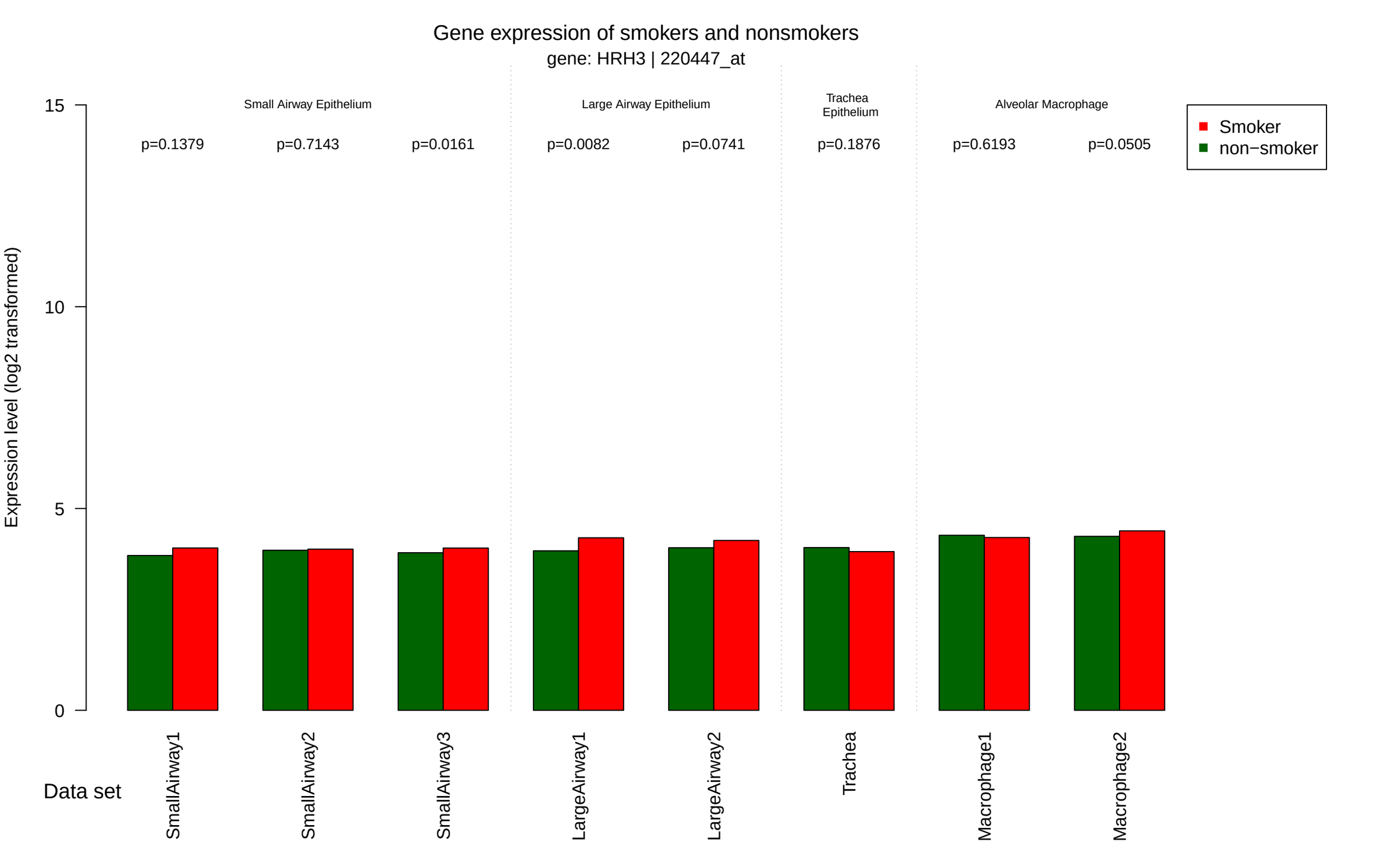
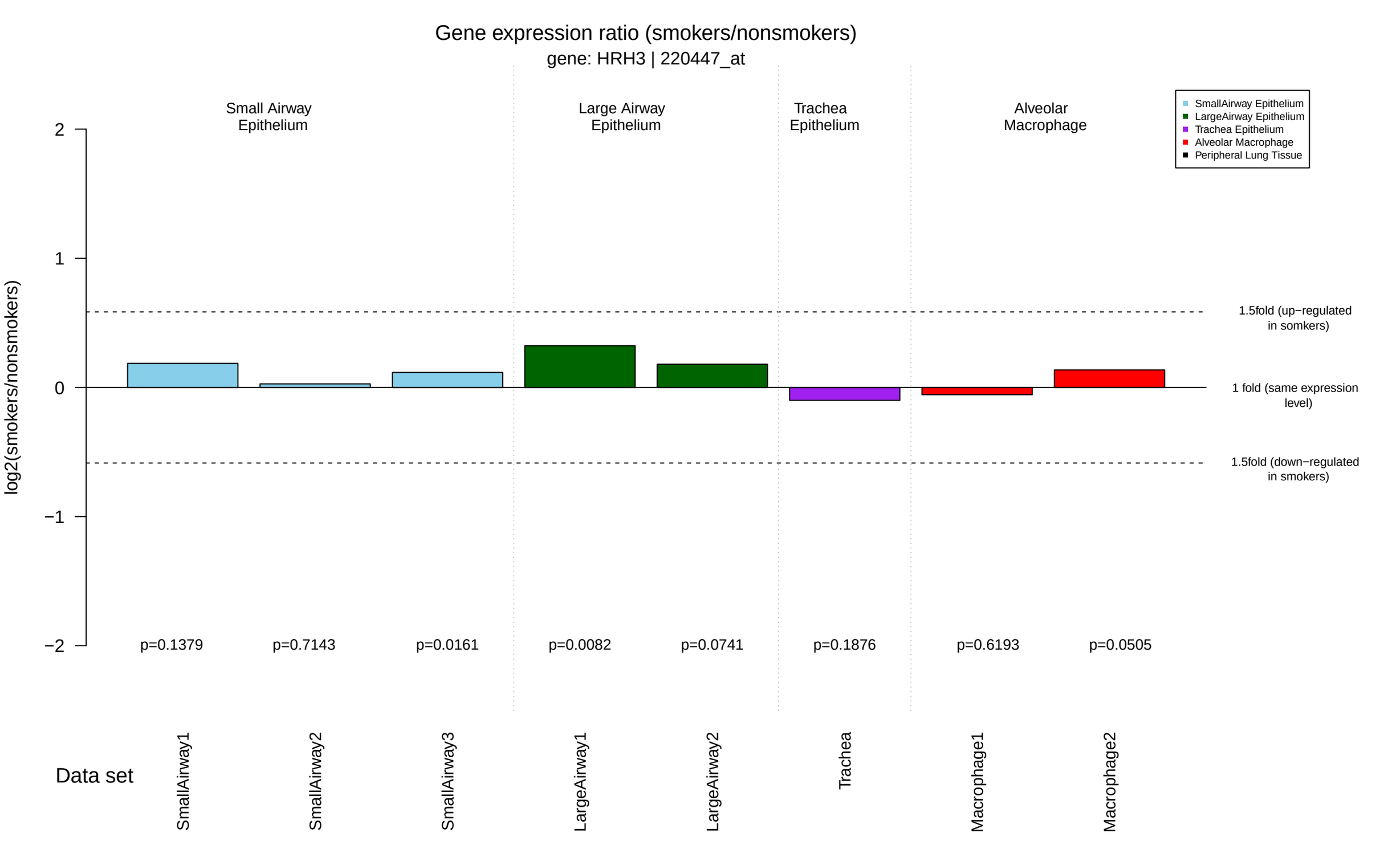
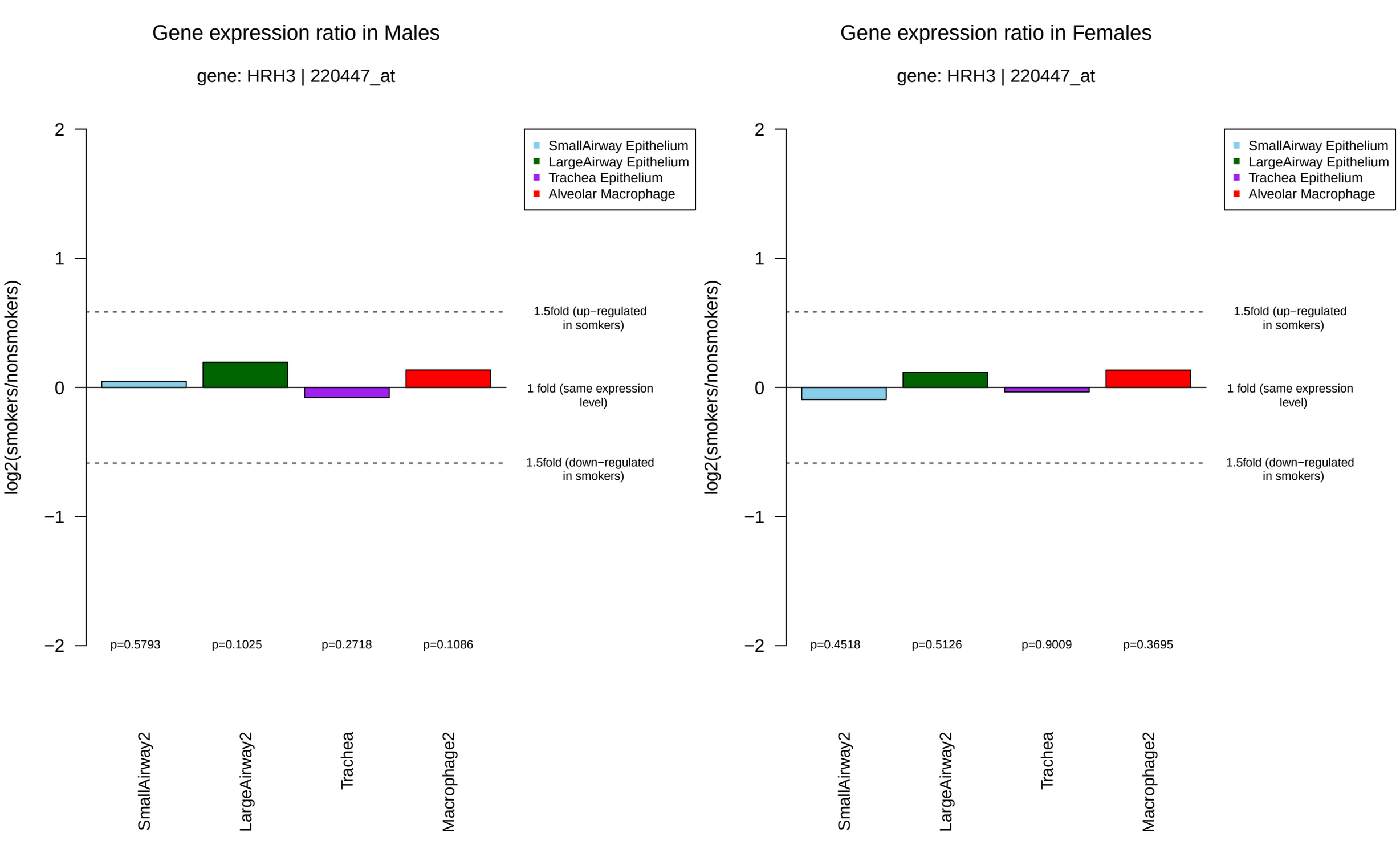
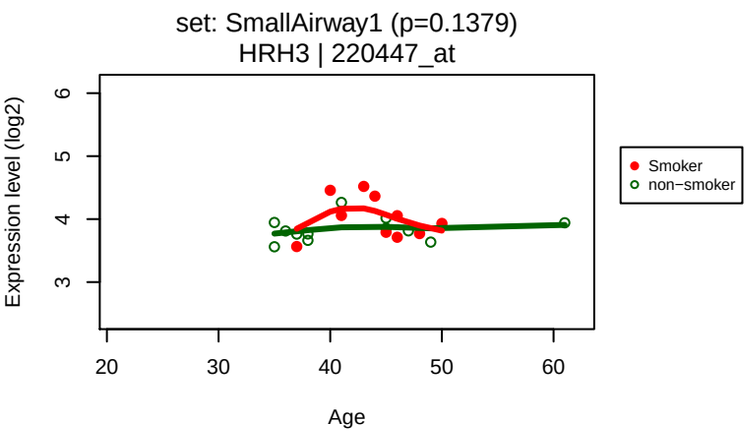
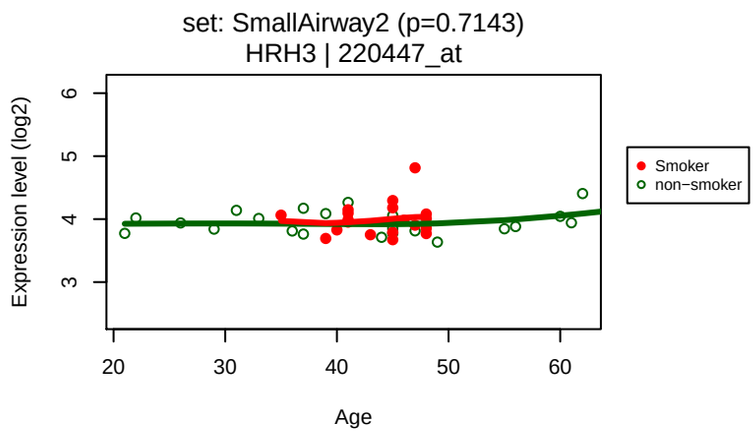
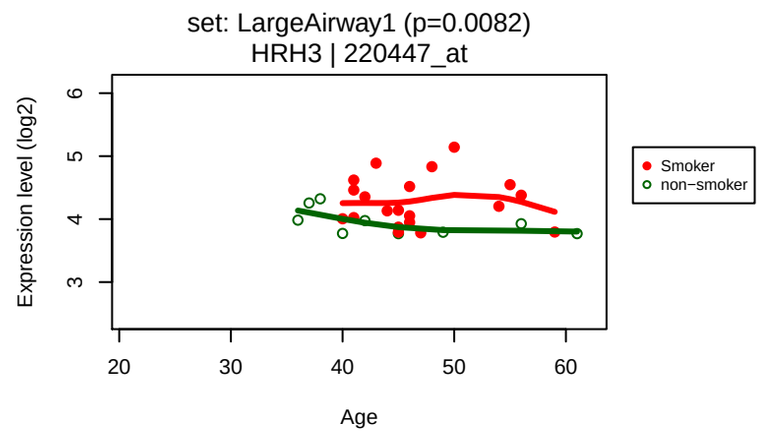

Gene expression of smokers and nonsmokers
gene: HRH3 | 220447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1379
p=0.7143
p=0.0161
p=0.0082
p=0.0741
p=0.1876
p=0.6193
p=0.0505
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

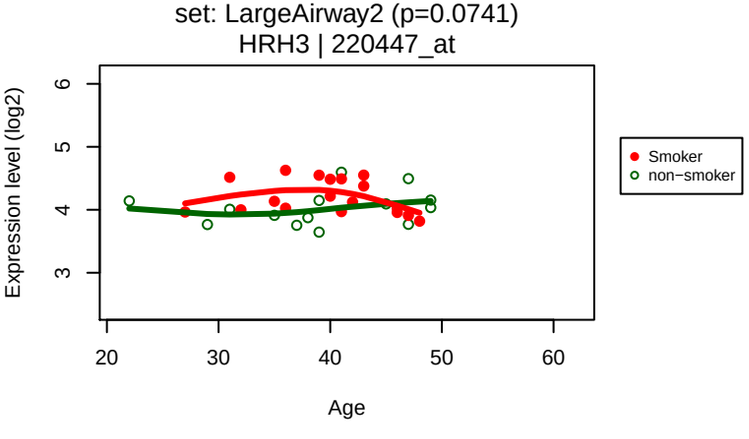

Gene expression ratio (smokers/nonsmokers)
gene: HRH3 | 220447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1379
p=0.7143
p=0.0161
p=0.0082
p=0.0741
p=0.1876
p=0.6193
p=0.0505
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

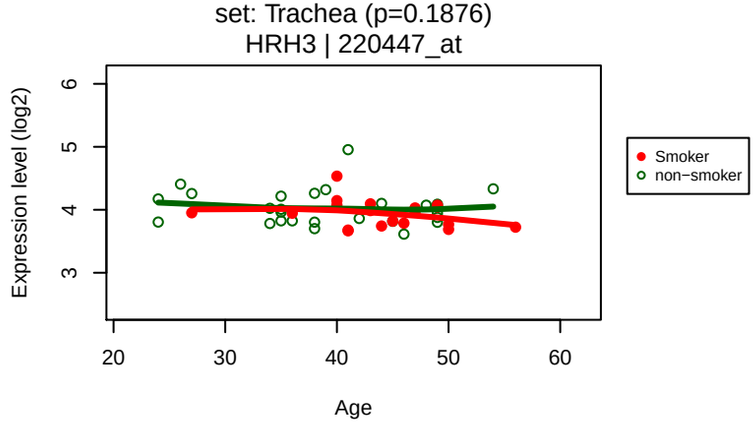

Gene expression ratio in Males
gene: HRH3 | 220447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5793
p=0.1025
p=0.2718
p=0.1086
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRH3 | 220447_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4518
p=0.5126
p=0.9009
p=0.3695
SmallAirway2
LargeAirway2
Trachea
Macrophage2

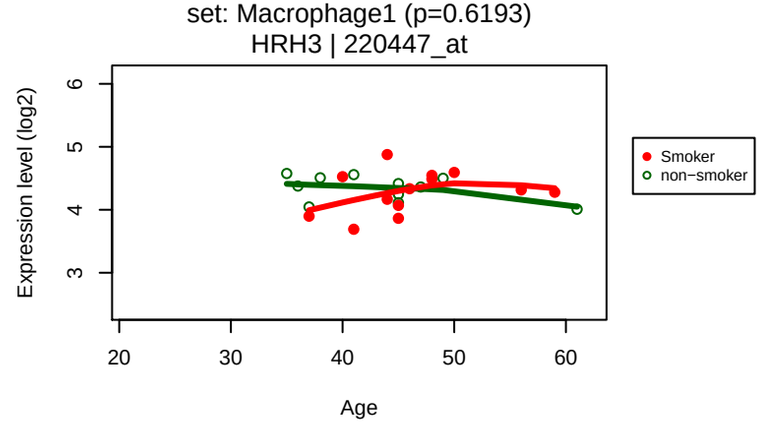

set: SmallAirway1 (p=0.1379)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker

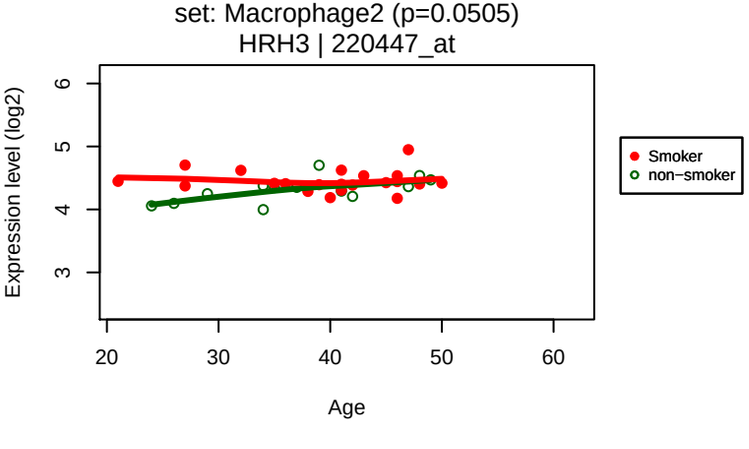

set: SmallAirway2 (p=0.7143)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker

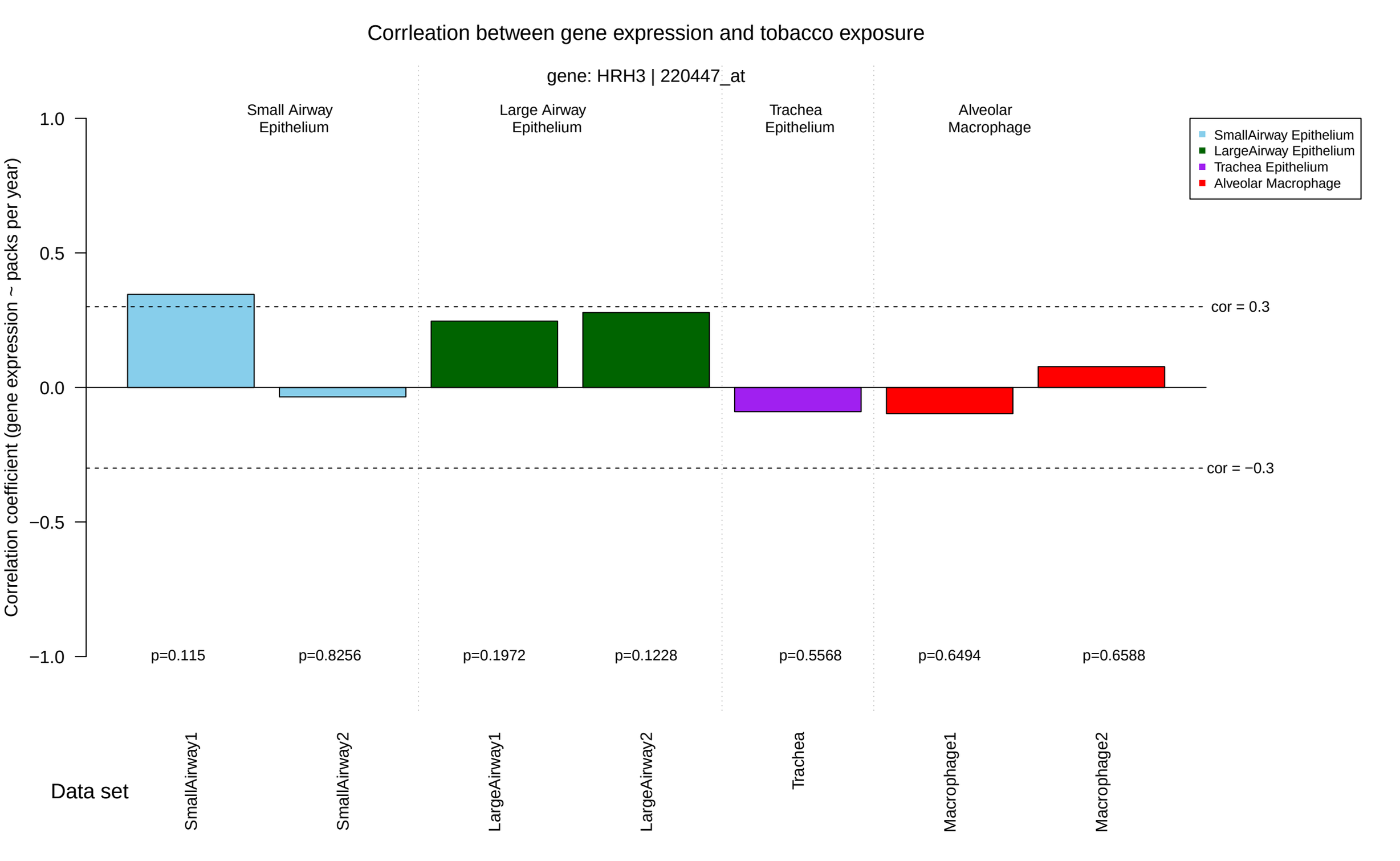

set: LargeAirway1 (p=0.0082)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0741)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1876)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6193)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0505)
HRH3 | 220447_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRH3 | 220447_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.115
p=0.8256
p=0.1972
p=0.1228
p=0.5568
p=0.6494
p=0.6588
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2